



SAVEETHA

INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
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ENGINEERING

DSA0101 – MS-1

Vehicle Service Center with Carbon Emission Monitoring System

|Object Oriented Programming with C++| SLOT-D|

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Abstract



A Vehicle Service Centre with Carbon Emission Monitoring System is a software application developed using C++ to simplify vehicle servicing operations while promoting environmental responsibility through emission monitoring. The system maintains customer records, vehicle information, service history, billing details, and appointment scheduling in a centralized platform. It also records carbon emission values obtained during vehicle inspection and compares them against predefined emission limits. If the measured emission exceeds the permissible level, the system alerts the service personnel and recommends maintenance activities such as engine tuning, oil replacement, air filter cleaning, or exhaust system inspection to reduce pollution. The application minimizes manual paperwork, improves data accuracy, and enhances service efficiency by automating routine operations. Different modules such as customer management, vehicle management, service management, emission monitoring, and report generation work together to provide a complete service management solution. The system can generate invoices, maintain historical service records, and produce emission reports that help both customers and service centre administrators monitor vehicle performance over time. By integrating environmental monitoring with regular vehicle servicing, the project encourages timely maintenance, reduces harmful emissions, and supports sustainable transportation practices. The application is designed with a user-friendly interface and structured programming concepts in C++, making it suitable for educational purposes as well as small and medium-scale automobile service centres. Overall, the project demonstrates how software technology can improve operational efficiency while contributing to environmental protection through continuous carbon emission monitoring and responsible vehicle maintenance.

Introduction



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Service Management Gaps

Manual records and scattered vehicle information. Diagnosis requires separate tools with limited customer visibility and difficult service history retrieval.



Environmental Impact

Poor engine condition leads to inefficient fuel usage and **increased emissions**. Pollutants — CO, CO₂, HC, NOx — from unserviced vehicles harm air quality.



Proposed Solution

A unified flow: **Vehicle** → **OBD Diagnostics** → **Service Management** → **Emission Monitoring** → **Dashboard** → **Alerts** — all in one integrated platform.

Literature Survey



S.No	Year	Author(s)	Method	Pros / Cons
1	2026	Patel et al.	AI-assisted diagnostics with emission monitoring	✓ Faster fault detection ✗ Higher implementation cost
2	2025	Kumar & Sharma	IoT-based vehicle service management	✓ Real-time monitoring ✗ Requires internet connectivity
3	2024	Singh et al.	Automated workshop management system	✓ Efficient scheduling & billing ✗ Limited environmental monitoring
4	2023	Rahman & Lee	Carbon emission tracking in automobiles	✓ Promotes pollution control ✗ Limited service integration

Existing Problems



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! Manual records are time-consuming and error-prone.

! Customer and vehicle information may become disorganized.

! Lack of centralized reports makes monitoring difficult.

! Difficult to retrieve previous service history quickly.

! Emission data is often maintained separately from service records.

! High-emission vehicles may not be easily identified for maintenance.

Objectives



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Smart Service Management

Centralized vehicle details, customer info, service records, and structured technician assignment with status tracking



Diagnostics & Emission Monitoring

Integrate **OBD-II data** for real-time vehicle condition monitoring; track emission levels and detect abnormal pollutant values



Intelligent Decision Support

Real-time dashboard, automated alerts, emission reports, and data-driven maintenance recommendations with historical analysis

Existing Systems Comparison



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System Type	Capabilities	Limitation
Traditional Service Center	Manual registration, paper records	No digitization or real-time insight
OBD Diagnostic Tools	Fault code reading, parameter logging	Diagnostics only; no service integration
Emission Monitoring Systems	Gas sensors, pollutant measurement	Standalone; not linked to service records
IoT Vehicle Monitoring	Sensors + cloud + dashboard	No service-center workflow support
Proposed System	All of the above — unified	Only integrated solution



Key Difference: Our system is the **only integrated solution** combining service management, diagnostics, and emission monitoring in one platform.

Proposed Solution



✓ Customer information management

✓ Service management

✓ Emission data recording

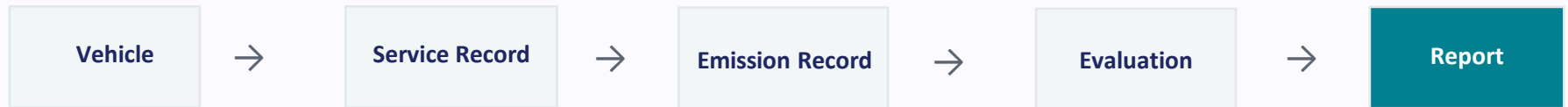
✓ Report generation

✓ Vehicle registration & tracking

✓ Service-history maintenance

✓ Emission-level evaluation

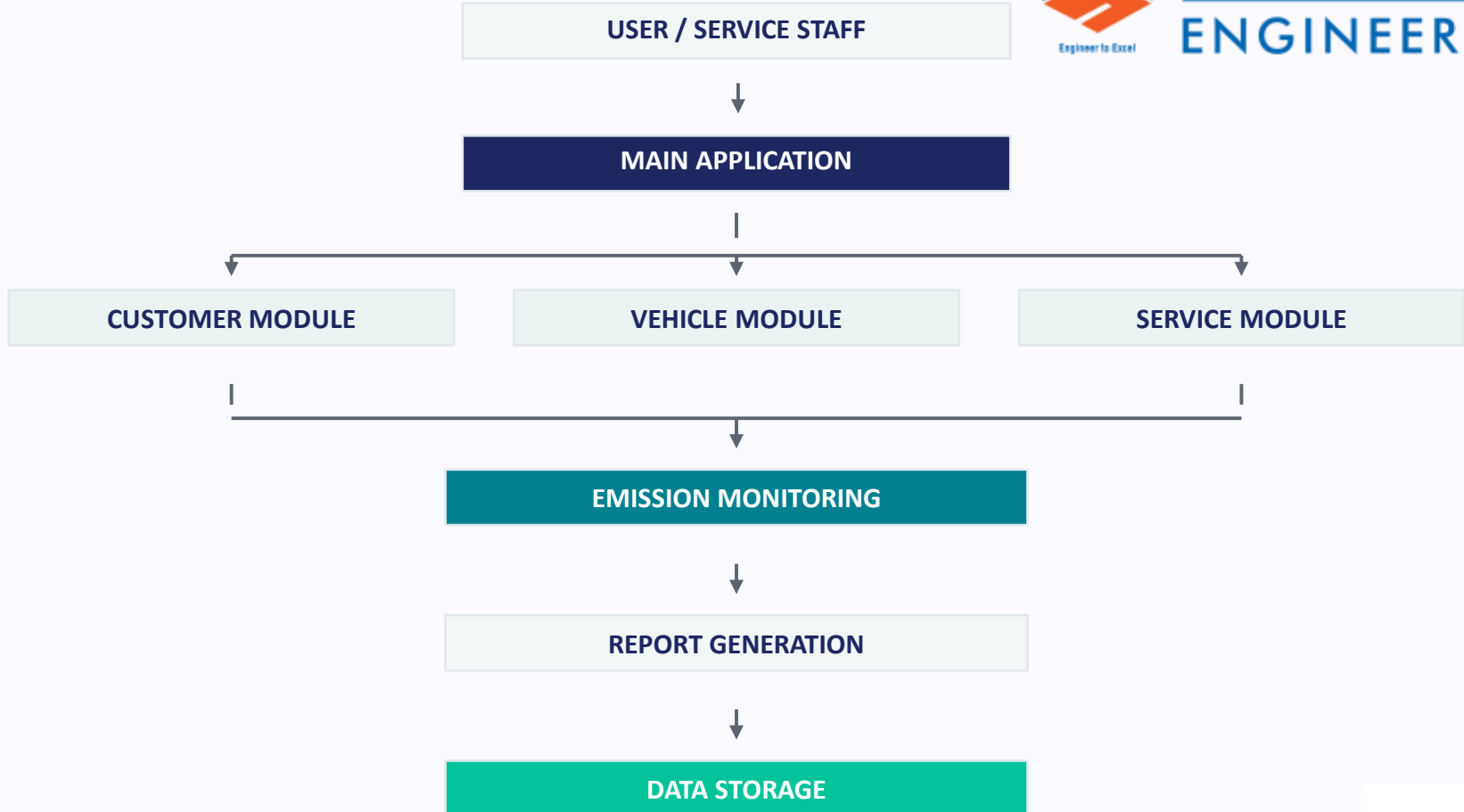
CORE IDEA



System Architecture



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Modules of the Project



Module 1 — Customer Management

Stores customer details and associated vehicle information in a centralized database for quick retrieval and updates.



Module 2 — Service Management

Maintains complete service history, manages appointments, and handles billing and invoice generation.



Module 3 — Carbon Emission Monitoring

Records emission values during inspection, compares them against standard limits, and triggers alerts when thresholds are exceeded.



Module 4 — Report Generation

Generates invoices, detailed service reports, and emission compliance reports for customers and administrators.

System Workflow



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START

1

Register Customer



2

Register Vehicle



3

Record Service
Details



4

Enter Emission Data



5

Evaluate Emission
Level



6

Save Service &
Emission Records



7

Generate Report



Technologies and Concepts Used

Technology Stack



Language

C++



Environment

[Code::Blocks / VS Code / Visual Studio]



Data Storage

File Handling

OOP Concepts Applied

- **Classes & Objects** — Represent customers, vehicles, services, emissions
- **Encapsulation** — Protects and manages class data
- **Inheritance** — Reuse of common attributes and functions
- **Polymorphism** — Functions behave differently per object
- **Abstraction** — Hides unnecessary implementation details
- **Constructors/Destructors** — Manage object init and cleanup
- **File Handling** — Stores and retrieves records

Advantages



Reduces dependence on manual records



Makes service-history tracking easier



Identifies potentially high-emission vehicles



Demonstrates real-world application of OOP



Provides centralized vehicle & customer information



Helps monitor vehicle emission levels



Generates organized reports



Expandable with databases, sensors, or online booking

Conclusion



- Provides an integrated approach to vehicle service management.
- Simplifies management of customers, vehicles, services, and emission records.
- Carbon emission monitoring adds an environmental aspect to traditional service-center management.
- Demonstrates practical implementation of Object-Oriented Programming using C++.
- Can be enhanced with real-time sensors, database connectivity, and automated notifications.

References



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THANK YOU

Vehicle Service Center with Carbon Emission Monitoring System